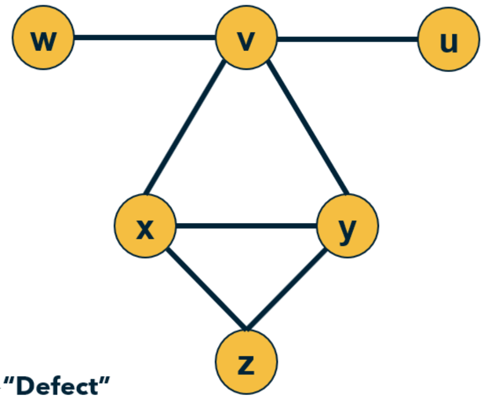
L10: Game Theoretic Diffusion Models

Player - 2

Player -1	Cooperate	Defect
Cooperate	b-c	-c
Defect	b	Ø



Initially all play “Cooperate” except u → “Defect”

Eventually all nodes play “Defect”

In some cases, the behavior of nodes in a social network can be captured more realistically with game-theoretic models: each node is a player that chooses rationally between a set of strategies. Crucially, the ***“payoff”*** of each strategy for player *v* also depends on the strategies chosen by the neighbors of *v*.

Let us illustrate this class of models with a simple coordination game. Each player can choose to either **“Cooperate”** *(work with others)* or **“Defect”** *(act selfishly).*

Cooperating with a neighboring player comes at a cost *c (independent of the strategy of that neighbor).* If two players cooperate, they both get a benefit *b* from their connection.

Of course, it only makes sense to cooperate if $b > c$.

The problem however is that some players may choose to Defect. In that case, they get the benefit b from every Cooperator they interact with – and they do not pay any cost. Think of the "friend" that will borrow things from you but he/she would never lend you anything.

The payoff matrix of this game for Player-1 is shown at the visualization.

Each player interacts only with its neighbors. Further, when a player chooses a strategy *(say to be a Cooperator)*, that strategy applies to all bilateral interactions with its neighbors.

So, if a player *v* has *k* connections, and *m* of those neighbors are Cooperators, then if *v* chooses to be a Cooperator its payoff will be $bm - ck$. It obtains a benefit b from each neighbor, but pays an additional penalty c for every connection.

If it chooses to be a Defector, its payoff will just be bm.

If a node is surrounded by Cooperators, then it would benefit in the short-term by becoming a Defector. But its neighbors would also decide to do the same, and they would all become Defectors.

So eventually, the payoff of all players will be lower than if they had all remained Cooperators.

This illustrates how selfish behavior may quickly spread on a network, even if it is harmful to everyone in the long term.

Food For Thought

The model we described here is very simple and it predicts that even a single Defector is sufficient to turn the whole network into Defectors.

More sophisticated models, based on evolutionary game theory, predict that under certain conditions that depend on the network topology, the strategy of Cooperation will persist even in the presence of some Defectors.

We recommend you read the paper: ***"* (https://www.nature.com/articles/nature04605) *A simple rule for the evolution of cooperation on graphs and social networks"*** (https://www.nature.com/articles/nature04605) by H. Ohtsuki et al, Nature, 2006.